

Masked-Face Staff Entry Recognition

Progress Report — Proof of Concept

Recognising individuals from partial or occluded face regions

1. Executive Summary

- ▶ Staff entry-control system that recognises individuals from partial or occluded face regions (surgical mask, hair cap).
- ▶ Current phase delivers a working proof-of-concept covering: face detection, embedding, open-set matching, staff enrollment, a live entry-monitoring camera, and an administrative dashboard for authorization management and audit logging.

Approved scope implemented and executable end-to-end



70%

- ▶ Remaining work: threshold calibration against real staff data, handling of concurrent unknown detections, and expanded field testing.

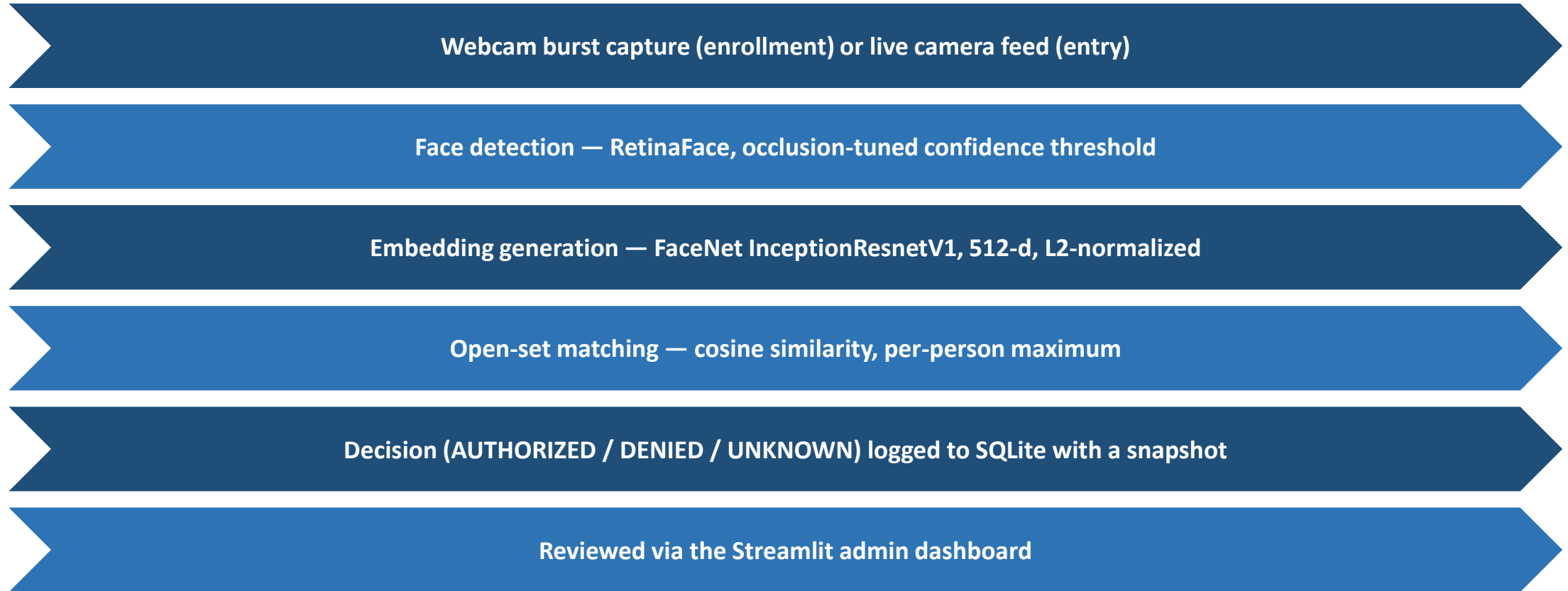
2. Implementation and Functional Progress

Modules Delivered

Module	File	Function
Detection + Embedding + Matching	face_engine.py	Core recognition logic: RetinaFace detection, FaceNet embedding, cosine-similarity open-set matching
Storage layer	db.py, schema.sql	SQLite persistence for staff, embeddings, and entry logs (WAL mode)
Configuration	config.py	Centralized tunables: thresholds, camera settings, capture parameters
Enrollment capture	capture_utils.py	Standalone webcam burst-capture window (8 photos), subprocess
Live entry monitor	entry_monitor.py	Real-time camera loop: detect, embed, match, draw decision, debounced logging
Admin dashboard	Home.py	Streamlit entrypoint, model warm-up, summary metrics
Staff registration UI	pages/1_Register_Staff.py	Guided burst-capture enrollment flow
Authorization management UI	pages/2_Manage_Authorization.py	List staff, toggle authorization, delete
Entry log viewer	pages/3_Entry_Logs.py	Filterable log table with decision counts and snapshot preview

Working modules representing ~50% of the approved project scope, all executable end-to-end.

2.2 System Architecture



2.3 Remaining Scope

- ▶ Calibration of the match threshold against real enrolled-staff data (tool built; awaiting sufficient enrolled staff).
- ▶ Handling of multiple concurrent unrecognized individuals (currently a single global cooldown bucket).
- ▶ Extended field testing across varied lighting, angles, and occlusion types.

3. Technical Accuracy and Use of Best Practices

Decision	Rationale
RetinaFace for detection	Selected specifically for robustness to facial occlusion (mask, cap), rather than a generic detector.
Detector confidence threshold lowered to 0.6	Occluded faces consistently scored below the typical 0.9 default, causing missed detections.
FaceNet (InceptionResnetV1, pretrained VGGFace2)	Carried forward from prior model comparison as the highest-performing embedding model.
Input pipeline: 160x160 resize, (pixel-127.5)/128.0 normalization, L2-normalized 512-d output	Matches facenet-pytorch's expected input specification exactly.
Open-set cosine-similarity matching (max similarity per person)	Allows new staff to be enrolled without retraining any model, unlike a closed-set classifier.
Background-thread detection worker	Keeps the live video preview rendering at camera frame rate while the slower, CPU-bound detect+embed pass runs asynchronously.
Centralized configuration module	All tunables (thresholds, paths, capture parameters) live in one place rather than as scattered magic numbers.

4.1 Interim Results — Baseline (Prior Research)

- ▶ Five model architectures evaluated for partial-face recognition on a custom dataset (423 original images, augmented to 2,202 via rotation, shear, zoom, horizontal flip, and brightness range adjustments).
- ▶ FaceNet achieved the strongest result and was carried forward into this implementation.

Metric	Value
Original dataset size	423 images
Augmented dataset size	2,202 images
Best-performing model	FaceNet (InceptionResnetV1)
Training accuracy	97.7%
Validation accuracy	85.5%

4.2–4.3 Current POC Configuration & Interim Metrics

Parameter	Value	Status
Embedding dimension	512	Fixed (FaceNet output)
Detector confidence threshold	0.6	Tuned for occlusion, empirically set
Minimum images to enroll	3 successful detections	Configured
Burst capture	8 photos at 1.2s interval	Configured

Interim System Metrics

- ▶ Entry-logs dashboard produces live, interpretable metrics for every recognition attempt: AUTHORIZED / DENIED / UNKNOWN counts, filterable by date and staff name, each with an associated snapshot image for audit and verification.
- ▶ Dedicated calibration tool computes genuine-pair vs. impostor-pair cosine-similarity statistics (min, mean, max, std) from enrolled staff embeddings, producing a data-driven threshold recommendation.
- ▶ Scheduled as the next data-collection milestone once a representative set of staff has been enrolled.

5. Problem-Solving and Technical Refinement

Occluded faces missed by the detector

Problem: Default RetinaFace confidence threshold (0.9) rejected masked/capped faces, which scored lower confidence than unoccluded faces.

Fix: Lowered detector confidence threshold to 0.6, restoring detection recall.

Streamlit's built-in camera input unsuitable for burst enrollment

Problem: `st.camera_input` captures only a single shot per rerun, too limited for an eight-photo enrollment burst with live feedback.

Fix: Built standalone OpenCV subprocess (`capture_utils.py`) with live preview and countdown.

Live video preview freezing during recognition

Problem: Synchronous `detect+embed` on every frame would block the video feed (CPU-bound, slow).

Fix: Moved detection/embedding to a background thread; main loop always renders latest completed result.

Open Issue — Identified, Not Yet Resolved

A single global cooldown bucket for unrecognized individuals means two different unknown people appearing in quick succession may only be logged once.

Documented as a known limitation — planned for the next iteration.

Thank You — Questions?